Robust Online Filter based on a Second-order Adaptive Model

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Abstract. This paper considers robust online filtering problems based on a second-order adaptive model. On the one hand, an adaptive second-order model is applied to characterize the nominal model of the data collected in real-time. On the other hand, we introduce a risk sensitivity parameter to express the modeling error between the nominal model and the actual model. More precisely, the proposed algorithm consists of two parts, one is the state estimation based on the robust Kalman filter, the other is the parameter identification using the Yule-Walker algorithm. Finally, results show the superiority of the proposed robust online filter.

Keywords: Online filtering, Robust Kalman filter, Modeling error, Yule-Walker algorithm

1 Introduction

With the improvement of the sensor technologies, people's requirements for data accuracy are getting higher and higher. However, data acquisition is always accompanied by noise, due to such reasons as measuring error, sensor fault, or loss of data. Hence, the key point is that how to reduce the influence of the noise and extract the useful information from the data set.

Currently, there are many algorithms focus on the off-line filtering problem. This means a full set of data is necessary before we deal with the data. About the off-line filtering, common solutions can be divided into two ways, one is translating the data set from time domain to frequency domain, the other is dealing with data set in time domain. More precisely, with respect to the off-line filtering in frequency domain, Bhati. D. et al. [1] design time-frequency localized three-band biorthogonal linear phase wavelet filter bank for epileptic seizure electroencephalograph (EEG) signal classification. Then, Kumar. M. K. P. et al. [2] propose a method for single channel speech separation (SCSS) by combining empirical mode decomposition (EMD) and speech specific information. In addition, there are also some off-line filtering which imposes the data to the time domain. For instance, Moon. T. et al. [3] present a new framework of applying deep neural networks (DNN) to devise a universal discrete denoise. Yang. X. et al. [4] presents a new model based on linear diffusion equation and noise estimation. However, these methods need all the data for the denoising process, and is not suitable for online purposes.

In addition, Kalman filter, as a well-known online filtering, has been widely used in a lot of applications. However, the performance of Kalman filter is highly dependent on the model precision. On the one hand, in piratical application of the online filtering, it is difficult to find an accuracy actual model which is equal to the nominal model. Hence, people propose many difficult statistic models, such as CV model and CA model [5]. These two models are suitable to some slightly fluctuating data. Then, in order to dispose the practical application, the Singer model [6] is proposed to deal with the colored process noise. Next, current model (CM) [6] is designed based on the Singer model and this model assumes that the gradient of data change at the next time tick is always within the neighborhood of the current value. Moreover, In order to better describe the dynamic characteristics of the system, Yi. S. L. et al. [7] propose a second-order adaptive statistic model to achieve the processing of the practical data set with colored process noise, and the characteristics of the colored noise were considered in the dynamic model via adaptive parameters.

On the other hand, even if we try to design a nominal model that is very close to the actual model, there is always a modelling error between the nominal model and the actual model. To address this, robust versions based on the standard Kalman filter was designed to guarantee some degree of immunity. Risk-sensitive filter [8] is the most critical one of these robust versions. In this approach, standard quadratic loss function is replaced by an exponential quadratic loss function. This exponential nature can penalize severely large errors. Furthermore, in [9], a ball is constructed by placing the bound on the Kullback-Leibler (KL) divergence between the the nominal probability density and the actual probability density and it represents an allowed mismodeling budget. Thus, a time-invariant risk sensitivity parameter over an entire simulation horizon is also introduced and larger parameter are able to penalize larger errors. Then, the Bayes estimator is robust under modeling errors.

In this paper, we apply an adaptive second-order model to characterise the nominal model of the data collected in real time. Moreover, in order to address the modelling error between the nominal model and the actual model, we introduce a risk sensitivity parameter. Then, the effectiveness of method was demonstrated via some experiments, which could have the best performance when the risk sensitively parameter is set as 0.005.

Finally, the structure of this paper is as follows. Section 2 presents the robust online filtering. The overview of the experiment is provided in Section 3. Some conclusions are given in Section 4.

2 Robust Online Filtering

We consider a nominal state space model of the form

$$\begin{aligned} x_{t+1} &= A_t x_t + B_t u_t + w_t \\ y_{t+1} &= C_{t+1} x_{t+1} + v_{t+1} \end{aligned} \tag{1}$$

where x_t, u_t and y_{t+1} are the state vector, the input vector and the measurement vector, respectively. A_t, B_t and C_{t+1} are the state process matrix, the input

matrix and measurement matrix, respectively. w_t is the process noise with its variance Q_t and v_{t+1} is the measurement noise with the known variance R.

2.1 State Estimation based on the Robust Kalman Filter

Here, we introduce the second-order adaptive model [7] to characterise the nominal model of the data collected in real time. Let $\mathbf{x}$ and $\dot{\mathbf{x}}$ denote the data itself and the gradient of the data, respectively. Then, the state vector should be $x_t := \left[\mathbf{x_t}\ \dot{\mathbf{x}}_\mathbf{t}\right]^T$. Next, the input vector is denoted by the mean of the estimated gradient of the data in the time interval of $[0, t]$, i.e.

$$u_t = \frac{1}{t}\sum_{k=0}^{t}\hat{\dot{\mathbf{x}}}_\mathbf{k}. \tag{2}$$

In addition, let $\hat{x}_{0|0}$, $u_0 = \hat{x}_{0|0}$ and $P_{0|0}$ be the initial value of the estimated state vector, input vector and covariance matrix, respectively. Moreover,

$$A_t = \begin{bmatrix} 1 & \frac{\left(1-e^{-\alpha_t T}\right)}{\alpha_t} \\ 0 & e^{-\alpha_t T}, \end{bmatrix}, \quad B_t = \begin{bmatrix} T - \frac{1-e^{-\alpha_t T}}{\alpha_t} \\ 1 - e^{-\alpha_t T} \end{bmatrix}. \tag{3}$$

Accordingly,

$$Q_t = 2\alpha_t\sigma_t^2 \begin{bmatrix} q_{11}\ q_{12} \\ q_{12}\ q_{22} \end{bmatrix} \tag{4}$$

where

$$\begin{aligned} q_{11} &= \tfrac{1}{2\alpha_t^3}\left(4e^{-\alpha_t T} - 3 - e^{-2\alpha_t T} + 2\alpha_t T\right) \\ q_{12} &= \tfrac{1}{2\alpha_t^2}\left(e^{-2\alpha_t T} + 1 - 2\alpha_t T\right) \\ q_{22} &= \tfrac{1}{2\alpha_t}\left(1 - e^{-2\alpha_t T}\right). \end{aligned} \tag{5}$$

where α_t and σ_t are self-adjusting parameters [11], T is the sample interval. In addition, we assume that C_{t+1} is a time-invariant matrix with $\left[0\ 1\right]$ and R is a known constant value.

Algorithm 1 robust Kalman filter

Require: y_t, $\hat{x}_{t|t}$, $P_{t|t}$

1: $\hat{x}_{t+1|t} = A_t\hat{x}_{t|t} + B_t u_t$
2: $P_{t+1|t} = A_t P_{t|t} A_t^T + Q_t$
3: $\tilde{P}_{t+1|t} = \left(P_{t+1|t}^{-1} - \lambda_t I\right)^{-1}$
4: $K_{t+1} = \tilde{P}_{t+1|t} C_{t+1}^T \left(C_{t+1}\tilde{P}_{t+1|t}C_{t+1}^T + R\right)^T$
5: $\hat{x}_{t+1|t+1} = \hat{x}_{t+1|t} + K_{t+1}\left(y_{t+1} - C_{t+1}\hat{x}_{t+1|t}\right)$
6: $P_{t+1|t+1} = \left(I - K_{t+1}C_{t+1}\right)P_{t+1|t}$

Since we obtain the state space model (1), it is not difficult to see that Kalman filter is commonly used to solve the state estimation problems of this model.

However, in the practical application, the actual model is always different from this nominal model. Therefore, a risk sensitivity parameter λ_t [10] is introduced to express the modelling error between the nominal model and the actual model. Note that if the risk sensitivity parameter is equal to 0, the robust Kalman filter is equivalent with the Kalman filter. Then, the implementation is given by Algorithm 1. Here, it is worth noting that this risk sensitivity parameter $\lambda_t \in (0, 1)$ is given by the expert experience. In other words, in different applications we have different empirical value which is calculated based on a large number of experiments. In addition, in our robust Kalman filter, there are two unknown parameters α_t and σ_t in A_t, B_t and Q_t. Accordingly, it is not difficult to see that if we want to know the estimated state $\hat{x}_{t+1|t+1}$, we must get α_t and σ_t at first.

2.2 Parameter Identification based on the Yule-Walker Algorithm

Then, we tend to find the self-adjusting parameters α_t and σ_t based on the Yule-Walker algorithm. Let α_0 and σ_0 denote the initial value of these two parameters. First, we define that $\hat{g}_t$ is the gradient of the data at time t and it is the second element of $\hat{x}_{t|t}$, i.e. $\hat{\dot{\mathbf{x}}}_{\mathbf{t}|\mathbf{t}}$. Then,

$$\begin{array}{l} r_t(1) = r_{t-1}(1) + \frac{1}{t}\left[\hat{g}_t\hat{g}_{t-1} - r_{t-1}(1)\right] \\ r_t(0) = r_{t-1}(0) + \frac{1}{t}\left[\hat{g}_t\hat{g}_t - r_{t-1}(0)\right] \end{array} \tag{6}$$

where $r_t(0)$ is the correlation function of the gradient between time t and time $t-1$, and $r_t(1)$ is the the autocorrelation function of the gradient at time t. Therefore, the parameters updating is given by [7]

$$\begin{array}{l} \alpha_{t+1} = -\frac{\ln r_t(1) - \ln r_t(0)}{T} \\ \sigma^2_{t+1} = \frac{r_t(0) - \alpha(t) r_t(1)}{1 - \left(\frac{r_t(1)}{r_t(0)}\right)^2}. \end{array} \tag{7}$$

In what follows, the equation (7) is applied to A_t, B_t and Q_t. Finally, we get a closed loop of the robust online filter as shown in Figure 1.

3 Experiment

The configuration of the experiment is shown in Figure 2. It was a quasi-static test for the column made by Chinese Grade 345 steel and C30 Grade concrete [12]. Through this experiment, deformation displacement at different time samples were obtained, which correspond to the measurements in the proposed algorithm. Then, via the off-line filtering based on the whole data, the reference data is not difficult to collect.

In what follows, we introduce the proposed robust online filter to solve this denoising problem. As shown in the first figure in Figure 3, it is not difficult to see that the reference is submerged by the serious noise, so it is very necessary to deal with this data set. On the basis of the expert experience, we set the risk sensitively parameter as 0.005. Then, the second figure in Figure 3 shows a very

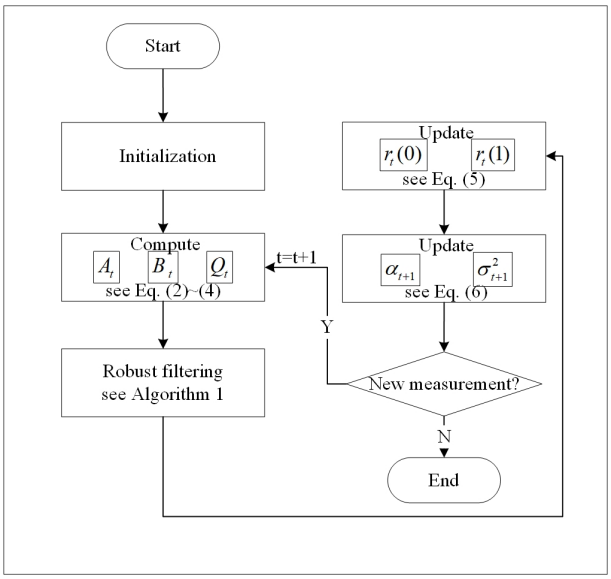


Fig. 1. The flow chart of the proposed robust online filtering algorithm.

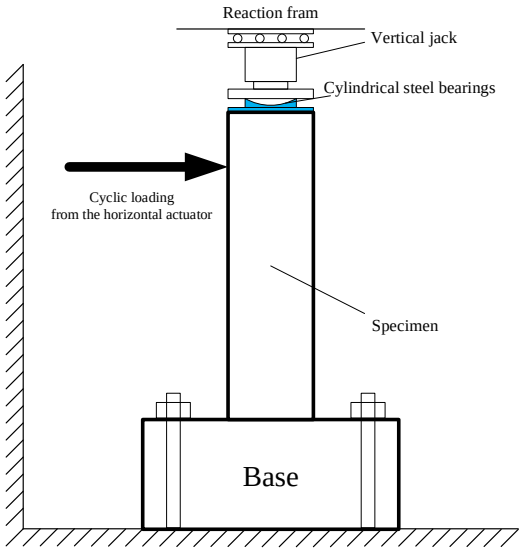


Fig. 2. The configuration of the experiment.

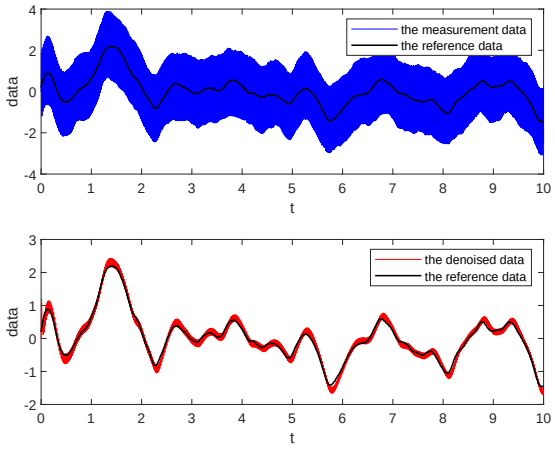


Fig. 3. First figure shows the measurement data and the reference data, and the second figure contains the reference data and the denoised data with the risk sensitively parameter $\lambda_t = 0.005$.

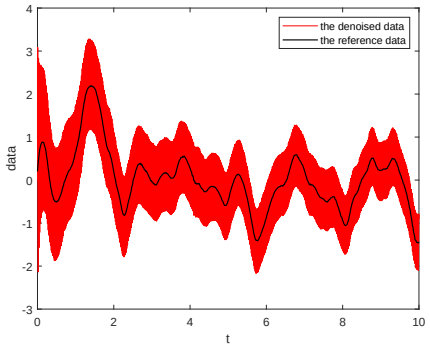


Fig. 4. The reference data and the denoised data with $\lambda_t = 0$.

satisfactory denoising effect. In order to better prove this conclusion, we select another five value of the risk sensitively parameter.

First, we set the risk sensitively parameter as 0. This means the proposed robust Kalman filter is equivalent with the Kalman filter. Comparing with the robust one with the standard one, it is clear to show in Figure 4 the performance of the robust online filtering is much better than the latter.

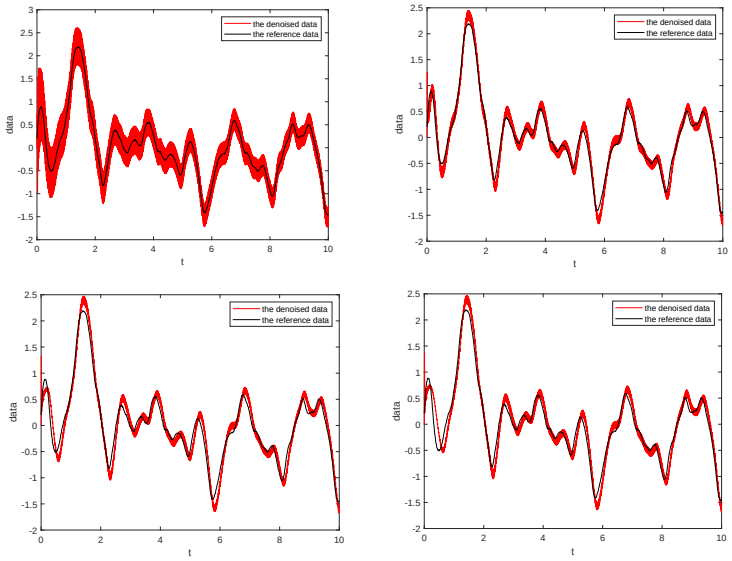


Fig. 5. The reference data and the denoised data with $\lambda_t = 0.001$, $\lambda_t = 0.01$, $\lambda_t = 0.05$, $\lambda_t = 0.1$, respectively.

Then, Figure 5 shows the performance of the reference data and the denoised data with the other different four risk sensitively parameters. In order to evaluate these results, the root-mean-square error (RMSE) is used. RMSE is very commonly used and makes for an excellent general purpose error metric for numerical predictions. More precisely, RMSE can be expressed as the following:

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{n} (d_i - r_i)^2}{n}} \tag{8}$$

where n is the number of the measurements, d_i is the ith denoised data and r_i is the corresponding reference data. Hence, Table 1 shows the performance comparison between different risk sensitively parameters. As shown in Table 1, it is not difficult to see that $\lambda_t = 0.005$ is the best choice in this application, and as the increase or the decrease of λ_t, the performance of the proposed algorithm would be worse. However, they all better than the situation of $\lambda_t = 0$. i.e. the standard Kalman filter.

Table 1. Performance comparison between different risk sensitively parameters.

λ_t	0	0.001	0.005	0.01	0.05	0.1
RMSE	0.6049	0.2310	0.1142	0.1275	0.1597	0.1684

4 Conclusion

In this paper, we propose a robust online filter based on the second-order adaptive model. Here, we focus on the practical case that the actual model is different from our nominal model, i.e. second-order adaptive model. Therefore, we introduce the risk sensitively parameter to express this modelling error. However, we only consider the expert experience of this risk sensitively parameter. Finally, the effectiveness of method was demonstrated via some experiments, which could have the best performance when the risk sensitively parameter is set as 0.005. In the future, we would consider to find the theoretical value of this risk sensitively parameter based on the minimax game [13].

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